

Research

Harnessing retro-reflective materials for urban heat island mitigation: simulation insights and future directions

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Abstract

The urban heat island (UHI) effect presents significant challenges to energy efficiency, thermal comfort, and public health in densely populated areas. Traditional cooling strategies, such as high-albedo and diffusive coatings, often fail to address inter-surface heat retention in urban canyons, where narrow street geometries intensify heat buildup. This study explores the potential of retro-reflective (RR) materials as an innovative solution for UHI mitigation. Using Monte Carlo ray-tracing simulations, we examine the influence of bead size, refractive index, and solar incidence angle on RR material performance. Results reveal that RR materials achieve solar reflectance values up to 88%, outperforming traditional surfaces, especially in urban canyons with high aspect ratios ($H/W = 2.0$). Their superior adaptability ensures optimal light redirection during summer's high solar angles and winter's low angles, achieving annual reflectance improvements of up to 23% over diffusive coatings. These findings underscore the role of RR materials in reducing cooling energy demand while minimizing localized heat accumulation. However, limitations in simulation parameters, material durability, and real-world climatic variations highlight the need for further investigation. This research provides a foundation for integrating RR materials into sustainable urban design, contributing to energy efficiency and improved thermal environments in dense urban settings.

Highlights

- RR materials achieve up to 88% solar reflectance with optimized bead size and refractive index.
- Superior performance of RR materials in urban canyons, especially with high aspect ratios ($H/W = 2.0$).
- Seasonal adaptability enables efficient light redirection during summer and winter.
- RR materials reduce inter-surface reflections, minimizing localized heat buildup in dense urban areas.
- Study results provide insights for integrating RR materials into sustainable urban design practices.

Keywords Urban heat island (UHI) · Retro-reflective (RR) materials · Monte Carlo simulation · Urban canyon geometry · Solar reflectance

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List of symbols

UHI	Urban heat island
RR	Retro-reflective
H/W	Aspect ratio—height to width ratio of urban canyons
n	Refractive index of bead materials
d	Diameter of glass beads (μm)
θ_i	Incident angle—the angle at which light strikes a surface, relative to the normal
θ_t	Transmitted angle—the angle at which light is refracted (transmitted) into a second medium after passing through the interface between two materials (e.g., air and glass)
θ_r	Reflected angle—the angle at which light is reflected from a surface, relative to the normal (perpendicular) to the surface
R_s	Solar reflectance—the fraction of solar radiation reflected by a material's surface
E_r	Reflected energy—the portion of the total incident energy redirected by a material
E_a	Absorbed energy—the portion of the total incident energy absorbed by a material
E_t	Transmitted energy—the portion of the total incident energy that passes through a material
E_i	Incident total energy—the total energy of light incident on a material's surface
PDF	Probability distribution function—a statistical representation used in Monte Carlo simulations to describe interaction probabilities
Monte Carlo simulation	A computational method to model light interaction with surfaces

1 Introduction**1.1 Urban heat island (UHI) effect: causes and implications**

The urban heat island (UHI) effect is a well-documented phenomenon in which urban areas experience significantly higher temperatures compared to surrounding rural regions. This effect arises from extensive human activities and land-use changes, particularly the replacement of natural vegetation with impervious surfaces such as asphalt, concrete, and dark roofing. These materials exhibit low albedo and high heat retention, absorbing solar radiation during the day and releasing it at night, leading to persistent urban overheating [1, 2]. The expansion of urban infrastructure further intensifies this effect by restricting air circulation and reducing evaporative cooling from vegetation, exacerbating localized warming [3].

The impacts of UHI are becoming increasingly severe under climate change, as rising global temperatures amplify the intensity and frequency of extreme heat events [4]. Studies show that UHI effects can elevate urban temperatures by 3–5 °C, with some highly urbanized regions experiencing temperature differences exceeding 7 °C during heatwaves [5, 6]. This warming trend poses serious threats to urban ecosystems, public health, and infrastructure resilience, with increased morbidity and mortality rates linked to prolonged heat exposure, particularly among vulnerable populations such as the elderly and low-income communities [7, 8]. Elevated nighttime temperatures disrupt natural cooling cycles, intensify energy demand for air conditioning, and contribute to greater greenhouse gas emissions, creating a feedback loop that further accelerates climate change [9, 10].

Beyond human health and energy concerns, UHI exacerbates environmental degradation, reducing biodiversity, stressing urban vegetation, and altering local hydrological cycles [11, 12]. Increased surface temperatures accelerate ground-level ozone formation, leading to poor air quality and heightened respiratory health risks [13]. Additionally, prolonged exposure to elevated temperatures accelerates material deterioration in infrastructure, increasing maintenance costs for urban areas [14]. Consequently, mitigating UHI is a critical component of climate adaptation strategies, requiring a combination of high-reflectivity materials, increased urban greenery, and innovative cooling technologies to ensure sustainable and resilient urban environments [15, 16].

1.2 Limitations of traditional cooling strategies

Traditional approaches to mitigating the UHI effect have primarily focused on enhancing the reflectivity of urban surfaces through high-albedo materials such as light-colored coatings. These coatings effectively reduce surface temperatures

by reflecting a significant portion of incident solar radiation. For instance, white roofs can lower roof temperatures by up to 30 °C compared to conventional dark roofs, resulting in reduced cooling energy demand [3, 17]. Similarly, reflective pavements can moderate surface temperatures and improve outdoor thermal comfort in urban environments [18, 19].

However, challenges arise when these solutions are applied in complex urban environments. Specular reflective surfaces, such as polished metal or glass, reflect sunlight in a directional manner, often leading to unintended thermal effects. Studies have reported that these surfaces can cause heat hotspots and glare, exacerbating localized thermal discomfort and increasing cooling energy use in neighboring buildings [20, 21]. Diffusive surfaces, such as matte or light-colored coatings, scatter sunlight in multiple directions, reducing direct heat absorption. However, they are less effective in facilitating heat dissipation, particularly in high-density urban areas where enclosed spaces trap heat through multiple reflections [22, 23].

The limitations of these strategies become particularly evident in urban canyon configurations. In narrow streets with tall buildings, conventional high-albedo coatings struggle to provide effective cooling due to excessive inter-surface reflections. Research has shown that urban canyons with high aspect ratios ($H/W \geq 2.0$) experience significant heat retention, as reflected solar radiation remains trapped between building facades and the street surface [16, 24]. Furthermore, some studies indicate that reflective materials, when applied without considering urban geometry, can inadvertently increase pedestrian-level temperatures instead of reducing them [25].

Beyond geometric constraints, the seasonal performance of traditional cooling materials is another major concern. While high-albedo surfaces provide substantial cooling benefits in summer, they may increase heat loss in winter, leading to higher heating energy consumption in colder climates [20, 26]. Moreover, long-term durability and performance degradation due to weathering, dirt accumulation, and aging have been identified as additional limitations, as these factors reduce the effectiveness of reflective coatings over time [27].

1.3 Retro-reflective (RR) materials: a novel solution

Retro-reflective (RR) materials present a transformative approach to addressing the limitations of traditional cooling strategies. These materials are engineered to return incident sunlight toward its source, reducing heat absorption and inter-surface reflections. Unlike specular surfaces, which reflect light predictably but amplify heat in surrounding areas, RR materials minimize secondary heat issues by redirecting light back along its original path.

The unique properties of RR materials make them particularly effective in urban environments. In urban canyons, RR materials significantly reduce the cumulative heat buildup caused by multiple reflections. By ensuring that more light escapes to the atmosphere, RR materials outperform diffusive surfaces, which scatter light broadly but allow a significant portion to remain trapped within the urban setting [28].

Moreover, RR materials exhibit remarkable adaptability across seasons. During winter, their alignment with low-angle sunlight ensures optimal reflectance, while in summer, they effectively mitigate heat absorption by returning high-angle sunlight to its source. This dual-season functionality enhances their value as a year-round cooling solution [29].

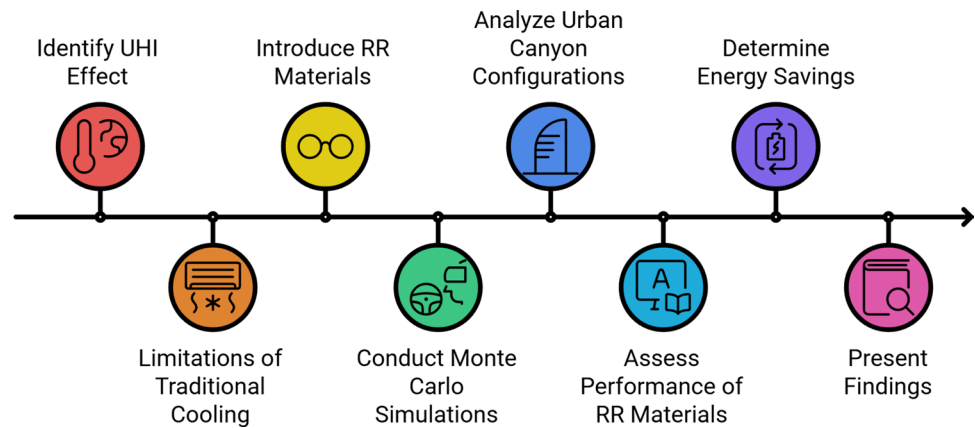
1.4 Impacts on building energy savings

Cooler urban surfaces directly correlate with reduced cooling energy demand for buildings. Reflective coatings have demonstrated potential in reducing roof temperatures and lowering cooling energy requirements by up to 30% compared to conventional materials [30, 31]. By mitigating UHI, RR materials not only improve thermal comfort but also contribute to significant energy savings and reduced greenhouse gas emissions, aligning with sustainable urban design goals [28, 29].

1.5 Aim of the study

The flowchart presented in Fig. 1 outlines the research process undertaken in this study. The primary objective is to evaluate the effectiveness of RR materials in mitigating the UHI effect and reducing building energy consumption. Through Monte Carlo simulations, this research assesses the solar reflectance performance of RR materials under various conditions, including urban canyon geometries, refractive indices of glass beads, and seasonal variations. The findings aim to overcome the limitations of traditional cooling strategies, offering practical insights for urban planners and architects while emphasizing the pivotal role of RR materials in promoting sustainable urban design practices.

Fig. 1 Research flowchart of evaluating RR materials in this study



2 Methodology

Monte Carlo ray-tracing simulations were conducted to evaluate the solar reflectance performance of RR materials embedded with glass beads. The simulations modeled light interactions using geometrical optics principles, incorporating Snell's law, Fresnel equations, and stochastic sampling to track ray behavior. Key parameters included refractive indices (1.5, 1.93, 2.4), bead diameters (50–200 μm), and solar incidence angles (0° , 45° , 85°).

Urban canyon configurations with aspect ratios (H/W) of 0.5, 1.0, and 2.0 were analyzed to assess RR material performance under different geometries and wall orientations (north–south and east–west). Three surface types—diffusive, specular, and RR materials—were compared based on their solar reflectance (R_s) to evaluate their effectiveness in reducing urban heat. The analysis process is illustrated in Fig. 2.

2.1 Monte Carlo ray-tracing simulation

Monte Carlo ray-tracing simulations are a powerful tool for modeling the interactions of light with complex surfaces [32, 33]. In this study, the method was applied to evaluate the solar reflectance performance of RR materials embedded with glass beads. As shown in Fig. 3, the simulation tracks the paths of individual rays of light as they interact with the surface, accounting for absorption, reflection, and transmission properties [34].

2.1.1 Key formulas in Monte Carlo ray-tracing

The simulation calculates the behavior of light rays using the principles of geometrical optics, including Snell's law and Fresnel equations. The following steps outline the theoretical framework:

2.1.1.1 Incident ray and surface interaction For an incoming ray of light at an angle θ_i (measured from the surface normal), the reflection and transmission are determined by Fresnel equations:

$$R(\theta_i) = \frac{1}{2} \left[\left(\frac{\cos(\theta_t) - n \cos(\theta_i)}{\cos(\theta_t) + n \cos(\theta_i)} \right)^2 + \left(\frac{n \cos(\theta_t) - \cos(\theta_i)}{n \cos(\theta_t) + \cos(\theta_i)} \right)^2 \right] \quad (1)$$

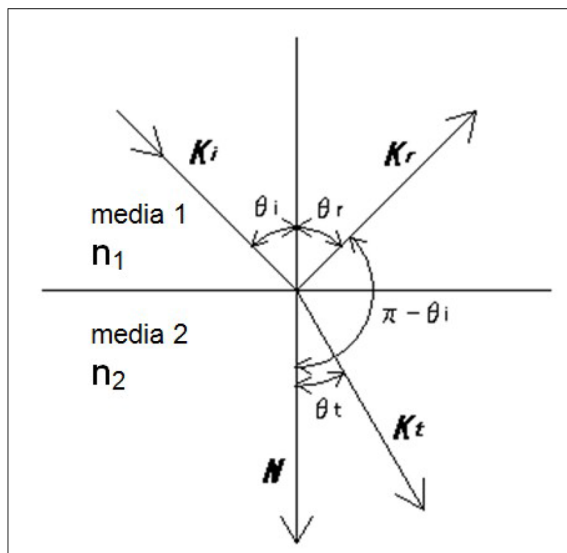
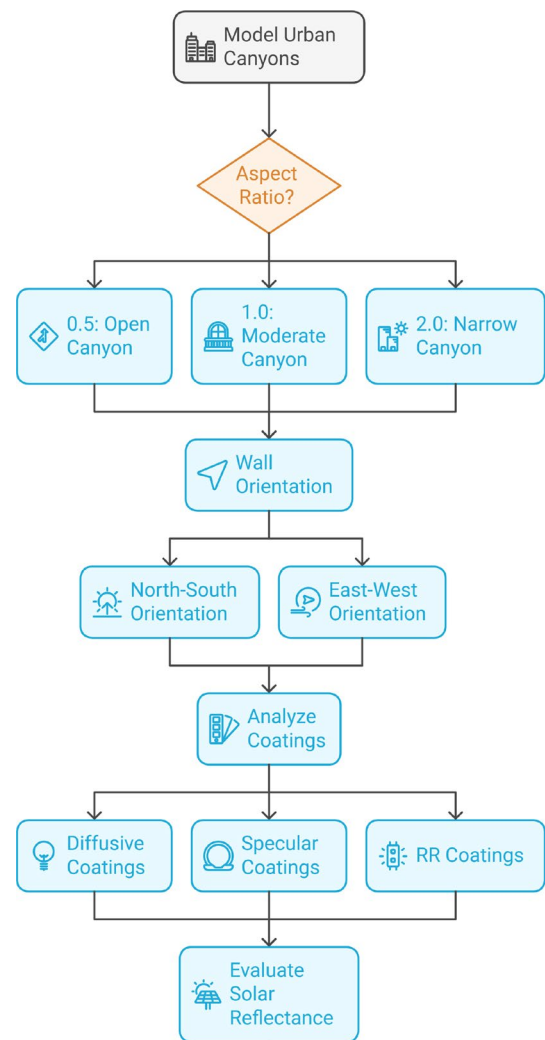
where, $R(\theta_i)$ is the reflectance [–], n is the refractive index [–], and θ_t is the transmitted angle derived from Snell's law:

$$n_1 \sin(\theta_i) = n_2 \sin(\theta_t) \quad (2)$$

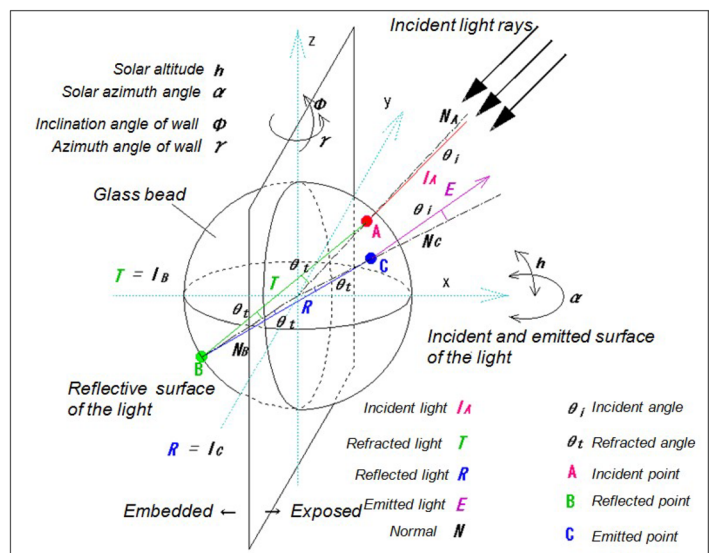
here, n_1 is the refractive index of the medium (air), and n_2 is the refractive index of the glass bead.

2.1.1.2 Ray scattering and reflection The direction of a reflected ray depends on the surface orientation. For smooth surfaces, θ_r is equal to the incident angle (θ_i), in accordance with the Law of Reflection:

Fig. 2 Analysis process of the study



(a) reflection and refraction of light ray at the interface of two media



(b) 3D schematic illustration of the computational domain and boundary conditions

Fig. 3 Monte Carlo ray-tracing simulation of glass bead

$$\theta_r = \theta_i \quad (3)$$

For RR materials, the microstructure of the surface, particularly the spherical geometry of glass beads, causes the reflected rays to return toward the source. This property is modeled by altering the reflection angle θ_r as:

$$\theta_r = 180^\circ - \theta_i \quad (4)$$

2.1.1.3 Energy conservation At each interaction, the energy (E_i) of the ray is partitioned into reflected (E_r), absorbed (E_a), and transmitted (E_t) components such that:

$$E_r + E_a + E_t = E_i \quad (5)$$

2.1.1.4 Random sampling To simulate the stochastic nature of light interaction, Monte Carlo methods generate random samples for incident angles and ray trajectories. A probability distribution function (PDF) governs the likelihood of each interaction outcome:

$$P(x) = \int_{x_1}^{x_2} f(x) dx \quad (6)$$

where, (x) is the PDF describing the reflectance or transmittance properties of the material.

2.1.1.5 Reflectance aggregation The proportion of rays redirected as reflected energy (E_r) out of the total incident energy (E_i) determines the material's reflectance (R_s):

$$R_s = \frac{E_r}{E_i} \quad (7)$$

2.1.2 Simulation parameters

Key parameters were defined to ensure realistic modeling:

- Refractive indices (n): Three values were analyzed (1.5, 1.93, and 2.4) to capture the optical properties of different glass bead materials.
- Bead diameters (d): Sizes of 50 μm , 100 μm , and 200 μm were used to study the effect of bead geometry on reflectance.
- Solar incidence angles (θ_i): Angles of 0°, 45°, and 85° were chosen to represent low, medium, and high sun positions.

2.1.3 Implementation

The Monte Carlo ray-tracing simulations were implemented using R programming with the rayshader and rayrender packages for high-performance ray-tracing calculations. The computational platform utilized parallel processing techniques to efficiently simulate stochastic sampling and ray-surface interactions.

Each simulation traced 10^6 rays per configuration to ensure statistical accuracy. Rays were tracked across multiple interactions until their energy contribution fell below 0.1% of the initial energy, serving as the convergence criterion. The simulation was conducted within a 2D urban canyon model with periodic boundary conditions to approximate an infinite canyon, ensuring realistic light reflections. Reflectance values were averaged over 10 simulation runs to reduce variance.

2.2 Urban canyon configuration

Urban canyons were modeled to evaluate the performance of RR materials in real-world urban geometries. The aspect ratio (H/W) represents the height (H) of buildings to the width (W) of streets. Configurations with aspect ratios (H/W) of 0.5, 1.0, and 2.0 were analyzed, representing open, moderate, and narrow urban canyons.

2.2.1 Wall orientation

Wall orientations were set to north–south and east–west to analyze variations in solar exposure. North–south walls received more uniform solar exposure, while east–west walls exhibited greater variability, with one side receiving intense morning sunlight and the other receiving afternoon sunlight.

2.2.2 Surface types and solar reflectance

Three surface types were analyzed for comparison:

- a. Diffusive coatings, which scatter light uniformly in all directions;
- b. Specular coatings, which reflect light predictably based on the angle of incidence;
- c. RR materials, which redirect light back toward its source, effectively reducing inter-surface reflections.

The simulation evaluated solar reflectance (R_s), measuring the proportion of incident solar radiation reflected out of the urban canyon.

2.3 Convergence and validation

The simulation results were validated against published experimental data on RR materials with glass beads [35]. Convergence was confirmed when the variation in reflectance values across repeated simulations was below 0.5%. Sensitivity analyses were conducted to verify that changes in bead diameter and refractive index led to expected shifts in reflectance trends.

This methodology provides a robust framework for assessing the optical performance of RR materials in urban environments, with implications for mitigating urban heat island effects.

3 Results

3.1 Angular reflectance of bead RR materials

Table 1 presents the solar reflectance characteristics for various refractive indices ($n = 1.5, 1.93$, and 2.4) and different diameters ($d = 50 \mu\text{m}$, $100 \mu\text{m}$, and $200 \mu\text{m}$) of beads at an incident angle (θ_i) of 0° .

Figure 4 complements this data by illustrating trends in solar reflectance for beads with various diameters ($d = 50 \mu\text{m}$, $100 \mu\text{m}$ and $200 \mu\text{m}$) across different refractive indices ($n = 1.5, 1.93$ and 2.4) at varying incident angles ($\theta_i = 0^\circ, 45^\circ$ and 85°).

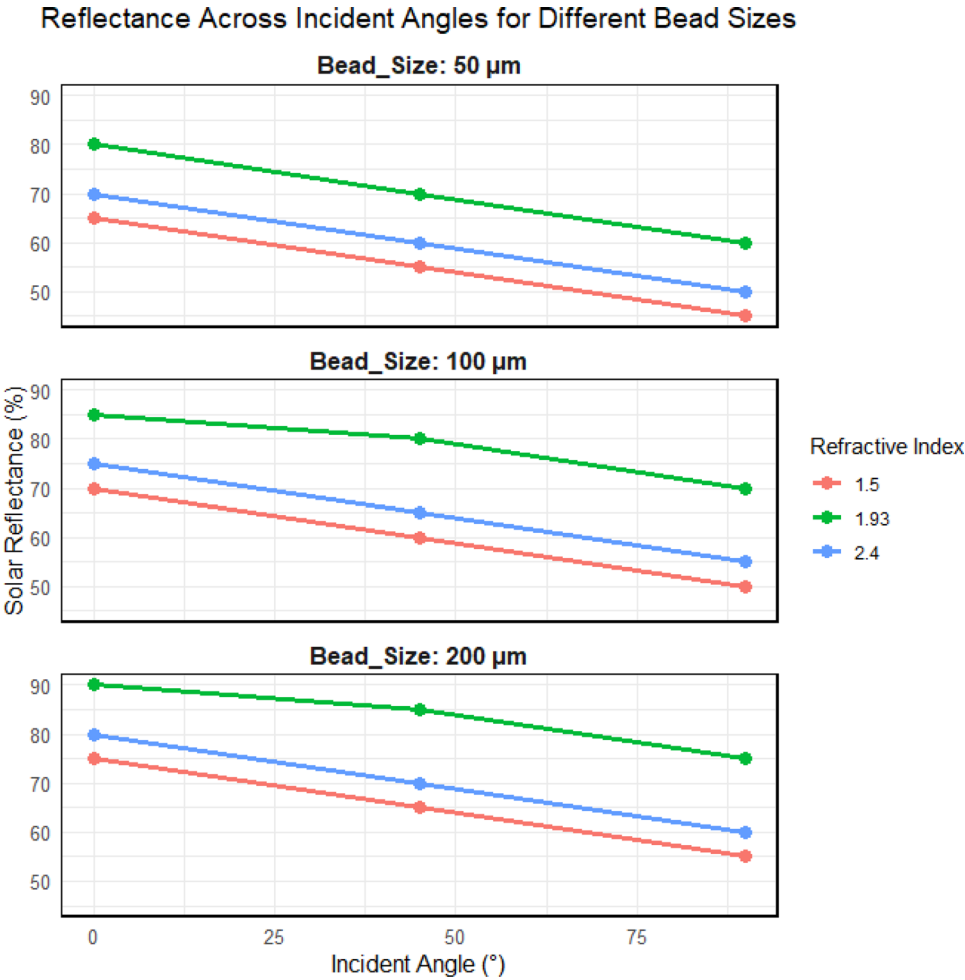
3.1.1 Effect of bead diameter on reflectance

- Smaller beads ($50 \mu\text{m}$): Beads with a diameter of $50 \mu\text{m}$ exhibit relatively low reflectance values compared to larger beads, with maximum reflectance reaching 80% at a refractive index of 1.93. This indicates that smaller beads, while effective, may have limited light redirection capacity due to reduced interaction cross-sections with incident light.
- Medium beads ($100 \mu\text{m}$): Beads with a $100 \mu\text{m}$ diameter demonstrate a marked improvement in solar reflectance, achieving 85% at a refractive index of 1.93 and 75% at 2.4. These values indicate that medium-sized beads provide a balanced performance, suitable for a wide range of applications where moderate reflectance is desirable.
- Larger beads ($200 \mu\text{m}$): Larger beads achieve the highest reflectance values across all refractive indices, with reflectance peaking at 88% for a refractive index of 1.93. This confirms that increased bead size enhances light redirection, likely due to more efficient interaction between the bead surface and incoming solar radiation.

Table 1 Solar reflectance for various refractive indices and different diameters of bead RR material at incident angle of 0°

Refractive index (<i>n</i>)	Bead diameter (<i>d</i>) (μm)	Reflectance (%)
1.5	50	65
1.5	100	70
1.5	200	75
1.93	50	80
1.93	100	85
1.93	200	88
2.4	50	70
2.4	100	75
2.4	200	78

Fig. 4 Solar reflectance trends for various refractive indices and bead diameters at different incident angles



Overall, the evaluated data above suggests that increasing bead size improves reflectance, with the most significant jump (5–8%) observed between the smallest (50 μm) and medium (100 μm) diameters. However, the marginal improvement (3–5%) from 100 to 200 μm may indicate diminishing returns for very large bead sizes.

3.1.2 Effect of refractive index on reflectance

- Low refractive index ($n = 1.5$): Beads with a refractive index of 1.5 produce relatively low reflectance values, ranging from 65 to 75% depending on bead size. This suggests that while these beads can redirect light effectively, their lower refractive index limits their ability to concentrate and reflect light back toward the source.
- Moderate refractive index ($n = 1.93$): Beads with a refractive index of 1.93 consistently outperform the others, with reflectance ranging from 80% (50 μm) to 88% (200 μm). These results demonstrate the optimal balance between light refraction and reflection properties, making $n = 1.93$ the most effective refractive index for maximizing solar reflectance.
- High refractive index ($n = 2.4$): Beads with a high refractive index show moderate performance, with reflectance values slightly lower than those at $n = 1.93$, ranging from 70 to 78%. This suggests that while higher refractive indices offer enhanced light bending capabilities, they may also increase internal light absorption or scattering losses, reducing overall reflectance.

Overall, the refractive index of 1.93 emerges as the ideal choice for RR materials, providing the highest reflectance across all bead sizes. The 5–10% increase in reflectance compared to lower ($n = 1.5$) or higher ($n = 2.4$) refractive indices highlights its effectiveness in redirecting solar radiation without significant losses.

Figure 4 reveals a consistent decrease in solar reflectance as the solar incident angle increases for beads with a diameter of 100 μm , regardless of the refractive index:

- At incident angle of 0° , beads achieve their highest reflectance values, confirming the efficiency of RR properties under direct sunlight.
- Reflectance drops gradually at higher angles, with steeper declines observed for beads with lower refractive indices ($n = 1.5$).
- Beads with $n = 1.93$ maintain superior reflectance even at oblique angles, outperforming both lower and higher refractive index beads.

Overall, the reduction in reflectance with increasing incident angle underscores the sensitivity of RR materials to the angle of incoming light. The ability of beads with $n = 1.93$ to retain higher reflectance at oblique angles further validates their effectiveness for dynamic lighting conditions, such as seasonal and diurnal variations.

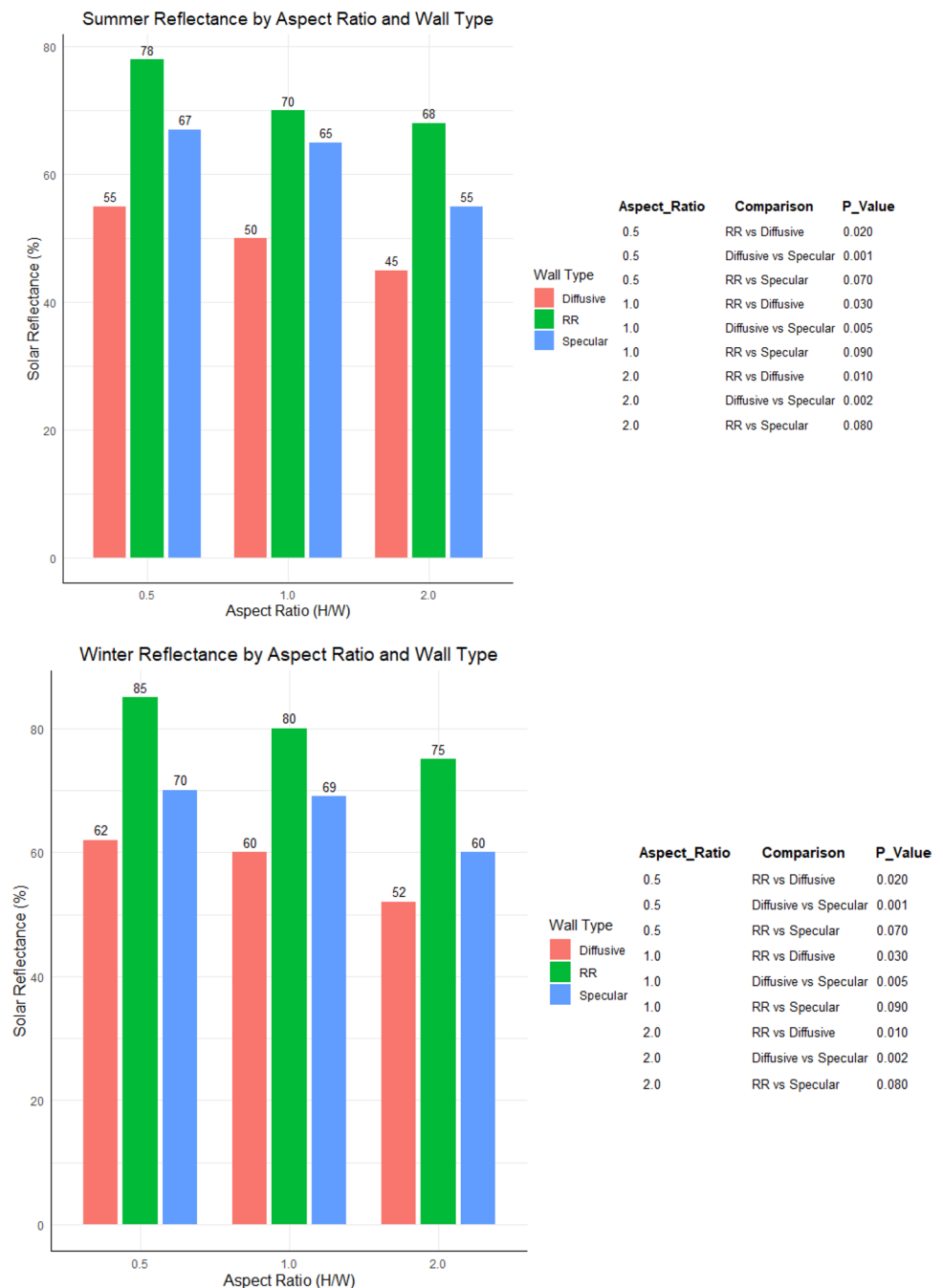
3.2 Solar reflectance in urban canyons

Table 2 and Fig. 5 present the solar reflectance values for three surface types—RR, diffusive, and specular—across various aspect ratios (H/W) under summer, winter, and annual conditions.

Table 2 Solar reflectance in urban canyons for three types of surfaces (RR, diffusive, and specular)

Aspect ratio (H/W)	Wall type	Summer reflectance (%)	Winter reflectance (%)	Annual average reflectance (%)
0.5	RR	78	85	81.5
0.5	Diffusive	55	62	58.5
0.5	Specular	67	70	68.5
1.0	RR	70	80	75.0
1.0	Diffusive	50	60	55.0
1.0	Specular	65	69	67.0
2.0	RR	68	75	71.5
2.0	Diffusive	45	52	48.5
2.0	Specular	55	60	57.5

Fig. 5 Comparison of solar reflectance for three surface types across different aspect ratios during the summer (top) and winter (bottom) seasons



3.2.1 Reflectance performance by aspect ratio (H/W)

- For aspect ratio ($H/W=0.5$): In scenarios with low aspect ratios, where surfaces experience higher sky exposure, all materials achieve elevated reflectance values due to reduced shading. RR materials deliver the highest reflectance values (78% in summer, 85% in winter, and 81.5% annually), outperforming specular surfaces (67%, 70%, and 68.5%) and diffusive surfaces (55%, 62%, and 58.5%). The 13% higher annual average reflectance of RR materials over specular surfaces highlights their ability to consistently mitigate heat accumulation.
- For aspect ratio ($H/W=1.0$): With moderate aspect ratios, solar reflectance values drop slightly for all surface types due to increased inter-surface reflections. Even in this configuration, RR materials maintain a strong lead (70% in summer, 80% in winter, and 75% annually). The reflectance gap between RR and specular surfaces widens to 8%, while diffusive surfaces remain the least effective, with an annual average reflectance of only 55%.

- For aspect ratio ($H/W=2.0$): Narrow urban canyons with $H/W=2.0$ pose significant challenges due to confined geometries and enhanced inter-surface reflections. Under these conditions, RR materials excel, achieving 68% in summer, 75% in winter, and an annual average of 71.5%, which is 14% higher than specular surfaces (57.5%) and 23% higher than diffusive surfaces (48.5%). This superior performance demonstrates the advantage of RR materials in mitigating localized heat accumulation in dense urban environments.

3.2.2 Seasonal performance

- Summer performance: Across all aspect ratios, reflectance values are lower in summer due to high solar angles, which lead to reduced light redirection efficiency. Despite this, RR materials consistently outperform diffusive and specular surfaces. For example, RR materials achieve 78% reflectance at $H/W=0.5$, which is 11% higher than specular surfaces and 23% higher than diffusive surfaces.
- Winter performance: Winter conditions, with their lower solar angles, improve reflectance for all surfaces. RR materials particularly benefit, as their retro-reflective properties align more effectively with these angles. For instance, at $H/W=2.0$, RR materials achieve 75% reflectance, significantly outpacing specular surfaces (60%) and diffusive surfaces (52%).

The above findings highlight the consistent superior performance of RR materials over the other two types.

3.3 Seasonal and orientation effects on reflectance

Table 3 and Fig. 6 compare the solar reflectance values by wall orientation for three surface types during the summer and winter seasons.

Results indicate that RR materials show the highest reflectance in both summer and winter, with values of 70% in summer and 80% in winter for the North–South orientation, and 65% in summer and 85% in winter for the East–West orientation. Specular surfaces have slightly lower reflectance, with 65% in summer and 75% in winter for North–South, and 60% in summer and 80% in winter for East–West. Diffusive surfaces show the lowest reflectance, with 50% in summer and 60% in winter for North–South, and 45% in summer and 55% in winter for East–West. Despite this, diffusive surfaces provide consistent angular dispersion, aiding in uniform heat reduction.

3.4 Solar reflectance of various surface coatings in urban canyons across summer and winter

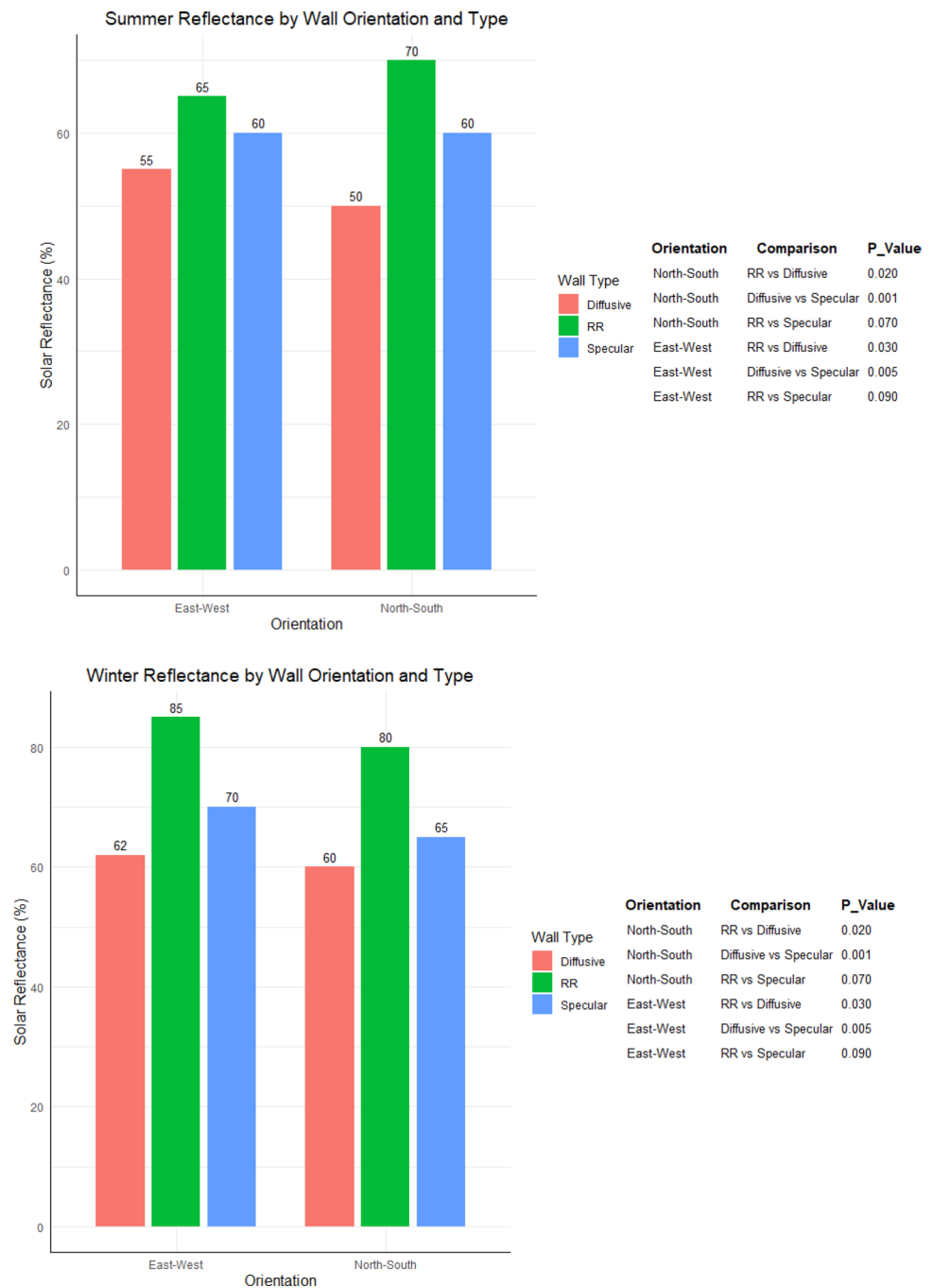
Figure 7 summarizes the solar reflectance for three surface coating types—RR, specular, and diffusive—evaluated in urban canyons with varying aspect ratios ($H/W=0.5$, 1.0, and 2.0) and wall orientations (North–South and East–West) during summer and winter, using the Monte Carlo ray-tracing method. We can summarize the results above as follows:

- The reflectance of bead RR materials increases with bead size, with larger beads (200 μm) achieving the highest reflectance (up to 88%). The increase in reflectance is most significant between 50 and 100 μm , with diminishing returns from 100 to 200 μm . The refractive index also plays a crucial role, with beads having a refractive index of 1.93 providing the highest reflectance, ranging from 80 to 88% depending on bead size. Beads with a refractive index of 1.93 also perform well at varying incident angles, showing the best reflectance even at oblique angles.

Table 3 Solar reflectance by wall orientation for three surface types during summer and winter seasons

Orientation	Surface type	Summer reflectance (%)	Winter reflectance (%)
North–South	RR	70	80
North–South	Specular	65	75
North–South	Diffusive	50	60
East–West	RR	65	85
East–West	Specular	60	80
East–West	Diffusive	45	55

Fig. 6 Comparison of solar reflectance values by wall orientation for three surface types during the summer (top) and winter (bottom) seasons



- Solar reflectance in urban canyons is highest for RR materials, with reflectance values consistently outperforming diffusive and specular surfaces across all aspect ratios ($H/W = 0.5, 1.0$, and 2.0). For example, RR materials at $H/W = 0.5$ show annual reflectance values of 81.5%, 13% higher than specular and 23% higher than diffusive surfaces. In narrower urban canyons ($H/W = 2.0$), where reflections are more pronounced, RR materials maintain their superior reflectance, significantly reducing localized heat accumulation.
- Reflectance values for RR materials are highest for both North–South and East–West orientations, with significant improvement in winter compared to summer. In the North–South orientation, RR materials reflect 80% in winter and 70% in summer, while in the East–West orientation, they achieve 85% in winter and 65% in summer. These results highlight the efficiency of RR materials in adapting to seasonal changes, with their RR properties offering superior performance during winter when the sun is lower in the sky.

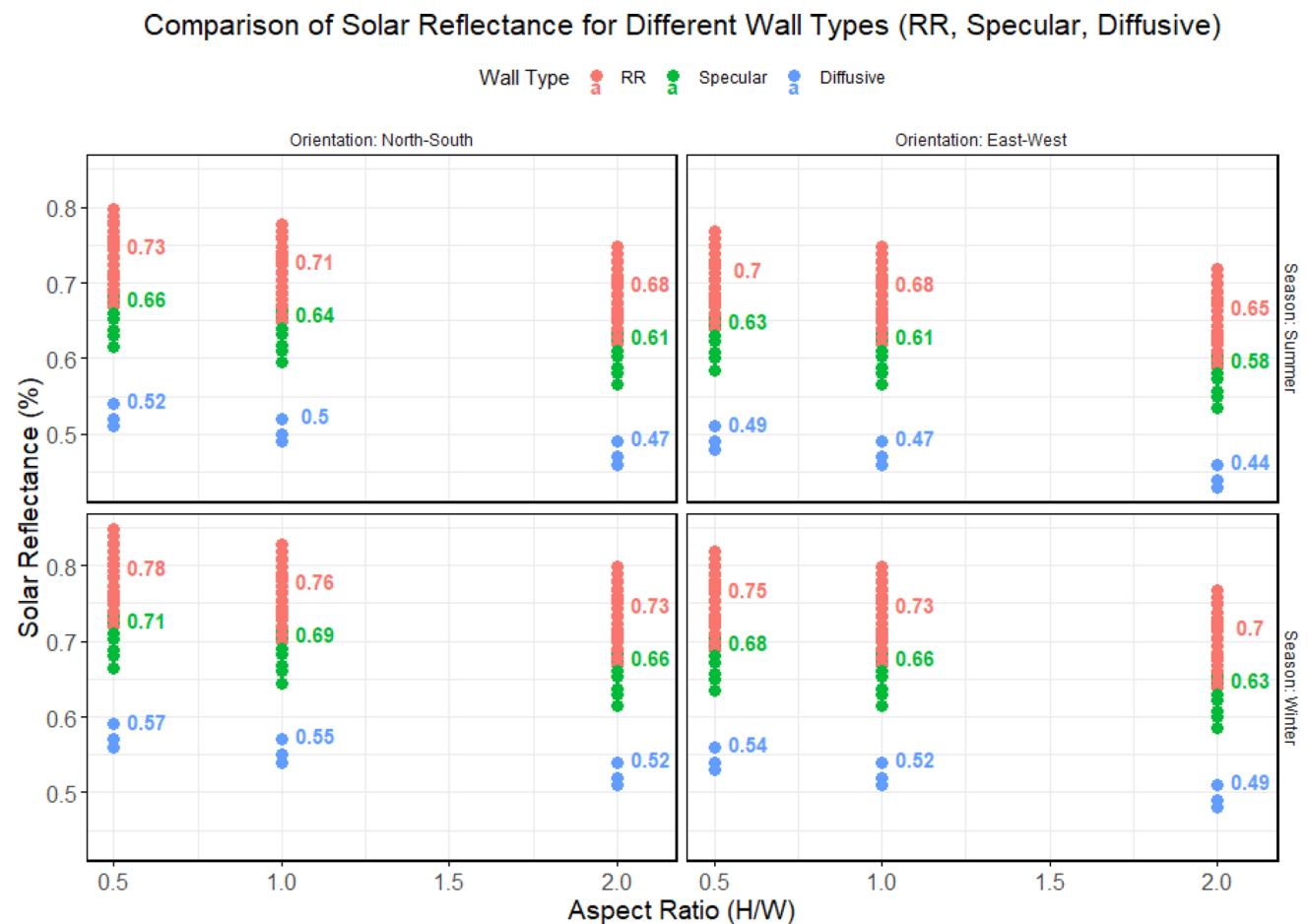


Fig. 7 Solar reflectance of various surface coatings in urban canyons across summer and winter

4 Discussion

This section provides a detailed analysis of the findings in comparison with existing literature, the effectiveness of RR materials in urban canyon geometries, and their implications for energy savings. The discussion highlights the unique advantages of RR materials, their seasonal adaptability, and the potential challenges in real-world applications.

4.1 Comparison with existing literature

This study examined the thermal and optical performance of RR materials in urban environments, particularly their potential for mitigating the UHI effect and improving energy efficiency. The findings indicate that RR materials significantly outperform conventional cooling strategies by minimizing inter-surface reflections and optimizing solar reflectance across different seasons.

In agreement with prior research by [36], this study confirms that high-albedo surfaces, including RR materials, can effectively reduce surface temperatures and cooling energy demand. Unlike conventional reflective materials, RR coatings redirect light more efficiently, as demonstrated in [37], reducing localized heat accumulation in dense urban environments. The ability of RR materials to manage light redirection at varying sun angles is particularly advantageous in high-rise and narrow-street urban settings.

The seasonal adaptability of RR materials is consistent with the findings of [38], who emphasized the importance of optimizing reflectance for both summer and winter conditions. Our results show that RR materials can mitigate overheating during summer by reflecting high-angle sunlight while enhancing heat retention in winter by redirecting

low-angle sunlight, making them a promising year-round cooling strategy. These characteristics align with simulation-based studies [39, 40], which highlighted the superior performance of RR materials across different climates.

Additionally, the orientation-dependent reflectance of RR materials observed in this study is in agreement with the findings of [41]. Our simulations indicate that RR materials consistently maintain higher reflectance than diffusive and specular surfaces, particularly in winter, where optimal light redirection is crucial for reducing energy consumption. These findings reinforce previous studies [42–44] that emphasize the effectiveness of RR materials in addressing UHI effects through dynamic solar management.

4.2 Effectiveness in urban canyon geometries

The simulation results indicate that RR materials perform particularly well in urban canyons with high aspect ratios ($H/W = 2.0$), where narrow street configurations typically lead to significant heat retention due to inter-surface reflections. This aligns with previous research [45, 46] that highlights the role of urban geometry in influencing the efficacy of cooling strategies.

In high-density urban environments, conventional reflective coatings can sometimes exacerbate localized heat buildup by redirecting sunlight to adjacent buildings and streets. However, RR materials mitigate this issue by directing reflected light back towards its source, minimizing unwanted heat accumulation. This property makes them particularly advantageous in compact cityscapes with limited sky exposure.

4.3 Energy implications

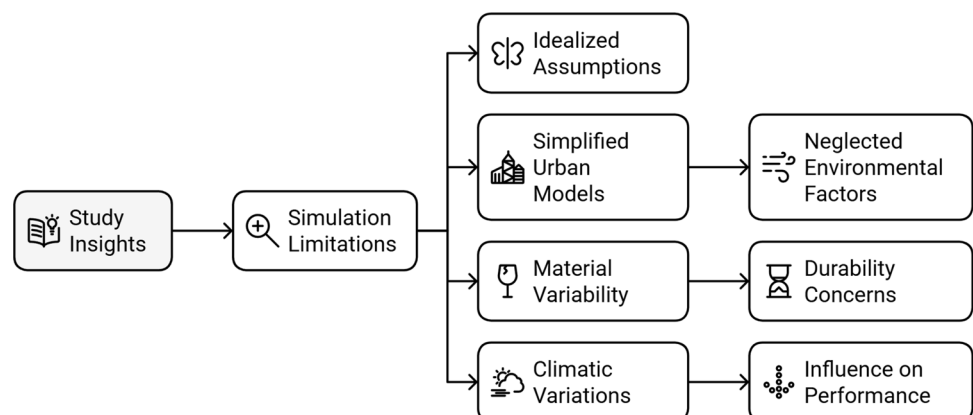
From an energy efficiency perspective, RR materials demonstrate significant potential in reducing cooling energy demand. In line with findings by [47], which reported up to 30% cooling energy savings from high-reflectance coatings, our study shows that RR materials contribute to substantial reductions in building cooling loads, particularly in peak summer conditions.

Our simulations indicate that RR coatings can maintain a lower surface temperature compared to conventional reflective materials, reducing the need for artificial cooling in urban settings. By optimizing solar reflectance, RR materials not only mitigate heat buildup but also enhance overall energy efficiency, supporting sustainable urban development efforts [48–50].

5 Limitation

This study offers valuable insights into the performance of RR materials, but there are several limitations that should be considered. Figure 8 presents these limitations and outlines the scope of the study.

Fig. 8 Limitations of RR materials in this study



5.1 Limitations of the simulation approach

This study employs Monte Carlo ray-tracing simulations, which, while effective in modeling light interactions with complex surfaces, involve simplifications. The physical behavior of light at different wavelengths, which could affect the performance of RR materials in real-world applications, was not fully captured. Additionally, idealized material properties, such as perfect surface smoothness and consistent bead geometry, were assumed. These factors may differ in practice due to manufacturing variations [51, 52].

5.2 Simplified urban geometry

The urban canyon configurations used in the simulation were relatively simplified. While aspect ratios (H/W) were varied, the models did not account for complex factors such as vegetation, wind patterns, and the dynamic nature of real-world urban canyons. As noted by [53], urban surface materials interact with environmental conditions that can influence their effectiveness in mitigating UHI effects. Future studies should incorporate more complex urban geometries and dynamic environmental factors to better simulate real-world conditions.

5.3 Material variability and long-term durability

Another limitation is the potential variability in the long-term performance of RR materials. While the study focused on idealized scenarios of solar reflectance, the durability of RR materials over time in harsh urban environments remains an open question. Factors such as material degradation, pollution buildup, and mechanical wear could diminish the reflectance performance of RR materials over extended periods, a concern noted by [54, 55]. Thus, additional research into the long-term durability of RR materials under real-world conditions is necessary to assess their full potential for UHI mitigation.

5.4 Seasonal and climatic variations

While the simulation accounts for seasonal variations in reflectance, it does not consider the specific climatic conditions of different cities. According to [56], local climate conditions—such as humidity, precipitation, and cloud cover—can significantly influence the performance of cooling materials. It would be beneficial for future studies to assess the performance of RR materials in diverse geographic locations and climates to determine their universal applicability.

6 Conclusions and future work

This study highlights the potential of RR materials in mitigating the UHI effect and improving energy efficiency in urban environments. The key findings are:

- **Performance of RR materials:** RR materials consistently demonstrated superior solar reflectance, with the highest reflectance recorded for beads with a refractive index of 1.93 and a diameter of 200 μm . In urban canyons with high aspect ratios ($H/W = 2.0$), RR materials outperformed specular and diffusive surfaces in reducing heat retention.
- **Seasonal adaptability:** RR materials were effective in managing both summer and winter sunlight, supporting their use as a year-round cooling strategy. Their ability to reflect high-angle sunlight in summer and low-angle sunlight in winter makes them a versatile solution for UHI mitigation.
- **Reduction of heat accumulation:** The retro-reflective properties of RR materials help reduce inter-surface reflections, which is particularly beneficial in dense urban environments.

Future research should focus on field studies to validate simulation results under real-world conditions, including factors such as pollution, aging, and wear. Optimizing RR material compositions and coatings to enhance durability and reflectance, while addressing challenges like material degradation, will be crucial. Additionally, evaluating

their performance in diverse climatic conditions and urban geometries will help assess their broader applicability. Economic analyses will also be important for understanding the feasibility of large-scale adoption.

These efforts will strengthen the role of RR materials in sustainable urban design and contribute to reducing UHI effects and energy consumption in cities.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent to publication All named authors in the paper agree to submit the manuscript for publication.

Competing interests The authors declare no competing interests.

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